

Quad-packing in the game EvenQuads

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Quad Packing Research Group

2024 Polymath Jr REU (virtual REU) (NSF award DMS-2218374)

- Mentored by:
 - ▶ Lauren Rose (Bard College)
 - ▶ Tim Goldberg (Lenoir-Rhyne University)
- Student researchers:
 - ▶ Taiki Aiba (Georgia Tech)
 - ▶ Dani Catalá (MIT)
 - ▶ and some others
- Presented at JMM 2025, MAA-SE 2025, AMS-SE S2025, and AMS-E F2024

Quad Packing Research Group, meets weekly on Zoom

- With folks above plus:
 - ▶ Hema Gopalakrishnan (Sacred Heart University)
 - ▶ Dyana Harrelson (Hope College)
 - ▶ Ren Watson (UT Austin)
- Presented at JMM 2026 and MAA-SE 2026

We were not the first!



[1] “Maximum Number of Quads”, 2024

- Student authors (grades 7-9): Byrapuram, Choi, A. Ge, S. Ge, ZiaLee, Liang, Mandal, Oki, Wu, and Yang - MIT PRIMES STEP Program
- Faculty author: Tanya Khovanova (MIT)

- Our methods are more visual and geometric.
- Had an important conjecture that we proved.

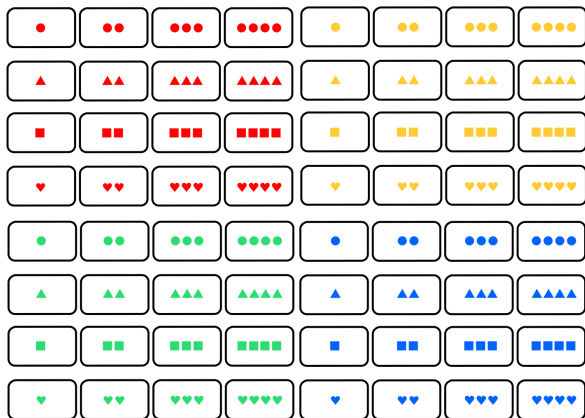
Origins of Quads

- *Quads*, initially called *SuperSET*, was introduced by Rose and Perreira in 2013.
- Published in 2021 as *EvenQuads* by the Association for Women in Mathematics in honor of its 50th anniversary. (Two different decks published so far, two more planned.)



The Quad-64 deck

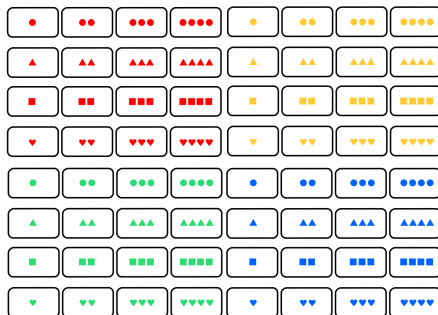
Each card has 3 attributes — color, shape, number — each with 4 possible values.



$$\#(\text{cards}) = 4 \times 4 \times 4 = 64.$$

Finding Quads

Goal: Find four cards that make a quad.



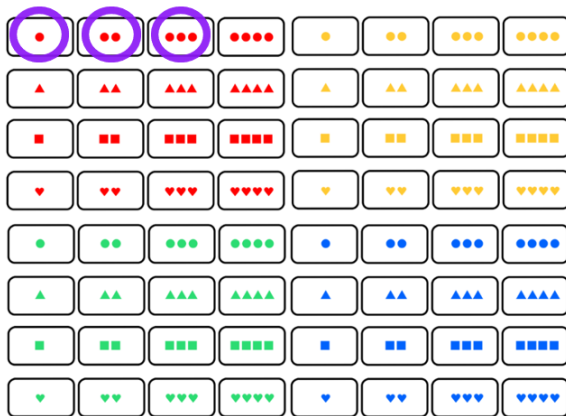
A **quad** is four cards where, for each attribute:

- 1 the values are all the same; or
- 2 the values are all different; or
- 3 the values are half-and-half.

Fundamental Theorem of Quads

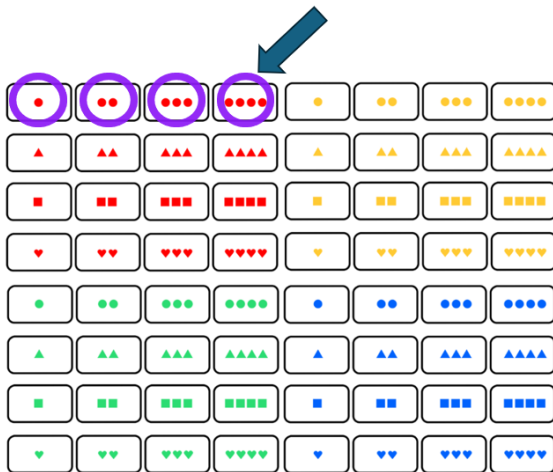
For any three distinct Quads cards, there is exactly one other card that forms a quad with them.

Some Examples of Quads



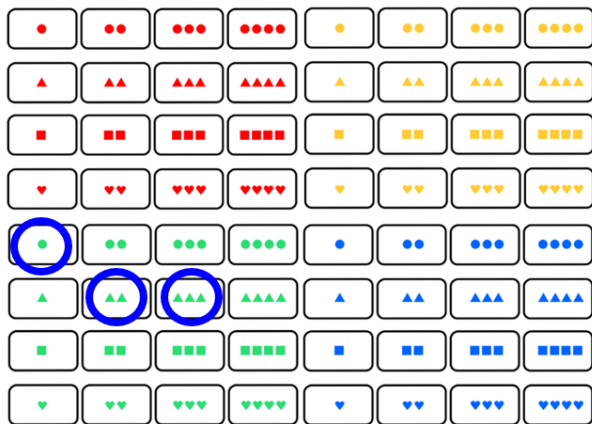
- different numbers, same color and shape

Some Examples of Quads



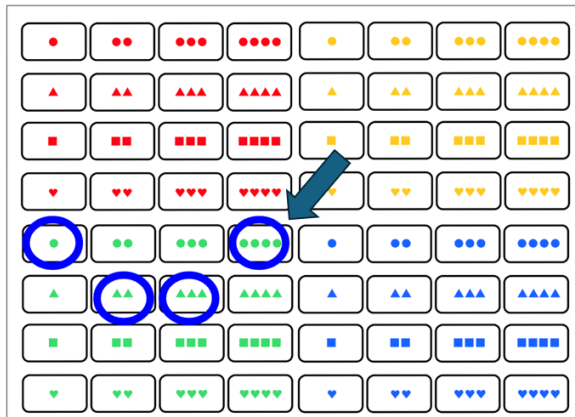
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Some Examples of Quads



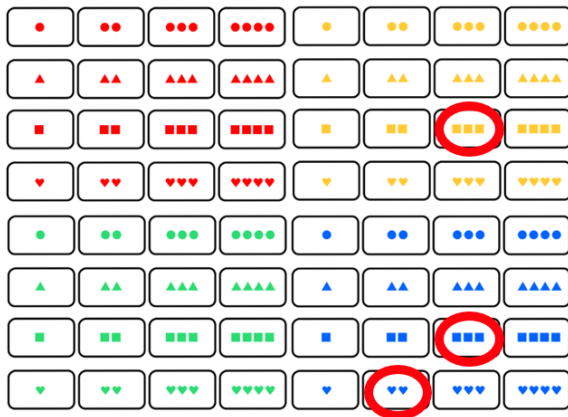
- different numbers, same color, half-and-half shapes

Some Examples of Quads



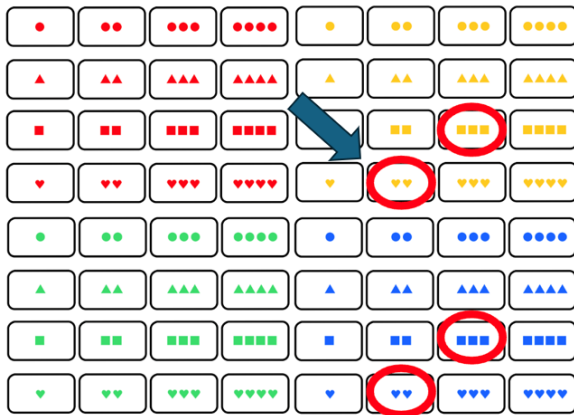
- different numbers, same color, half-and-half shapes

Some Examples of Quads



- half-and-half numbers, colors, and shapes

Some Examples of Quads



- half-and-half numbers, colors, and shapes

Coordinates for Quads Cards

Assigning coordinates to cards

- For each attribute, assign each value one of the elements of $\mathbb{Z}_2 \times \mathbb{Z}_2 = \{00, 01, 10, 11\}$.
- Then each Quads card is represented by an element of $(\mathbb{Z}_2 \times \mathbb{Z}_2)^3 \cong (\mathbb{Z}_2)^6$.

A possible assignment of coordinates:

Attribute	Values	Coordinates (respectively)
Color	red, green, yellow, blue	00, 01, 10, 11
Shape	circle, triangle, square, heart	00, 01, 10, 11
Number	one, two, three, four	00, 01, 10, 11

For example:

			
00 01 00	10 10 01	11 01 10	01 10 11

Possible Binary coordinates for Quad-64

 000000	 000001	 000010	 000011	 100000	 100001	 100010	 100011
 000100	 000101	 000110	 000111	 100100	 100101	 100110	 100111
 001000	 001001	 001010	 001011	 101000	 101001	 101010	 101011
 001100	 001101	 001110	 001111	 101100	 101101	 101110	 101111
 010000	 010001	 010010	 010011	 110000	 110001	 110010	 110011
 010100	 010101	 010110	 010111	 110100	 110101	 110110	 110111
 011000	 011001	 011010	 011011	 111000	 111001	 111010	 111011
 011100	 011101	 011110	 011111	 111100	 111101	 111110	 111111

Quads and Linear algebra

- The Quad-64 cards and game can be represented by the vector space $(\mathbb{Z}_2)^6$.
- For any positive integer n , there is a game **Quad- 2^n** . Its cards correspond to elements of $(\mathbb{Z}_2)^n$.

Theorem

Four cards form a quad if and only if the sum of their vectors is $\vec{0}$.



$$000100 + 101001 + 110110 + 011011 = 000000$$

Quads and Affine Geometry

Quad- 2^n cards are points of the affine space associated with $(\mathbb{Z}_2)^n$.

Theorem

*A set of points is a flat if and only if it is **quad-closed**, i.e. closed under triple sums of distinct points.*

- In general, an n -flat contains 2^n points.

- Lines (1-flats) are sets of two points.



- Planes (2-flats) are exactly the quads.



- The 3-flats (8 cards) are determined by any plane and one more point.



Maximum Number of Quads in ℓ Cards

Quad-Packing: What is the largest possible number, $M(\ell)$, of quads that ℓ cards can contain?

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- $\ell = 1, 2, 3$: $M(\ell) = 0$

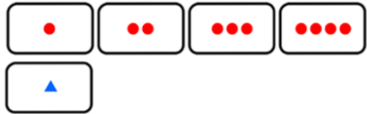
- $\ell = 4$:  $M(4) = 1$

Maximum Number of Quads in ℓ Cards

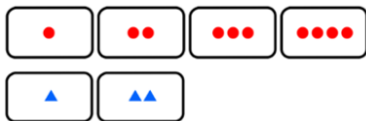
Quad-Packing: What is the largest possible number, $M(\ell)$, of quads that ℓ cards can contain?

- $\ell = 1, 2, 3$: $M(\ell) = 0$

- $\ell = 4$:  $M(4) = 1$

- $\ell = 5$:  $M(5) = 1$

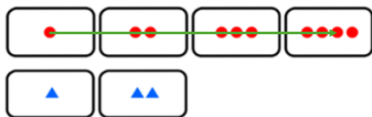
Maximum Number of Quads, $M(\ell)$, in ℓ Cards



• $\ell = 6$:

$$M(6) = 3$$

Maximum Number of Quads, $M(\ell)$, in ℓ Cards

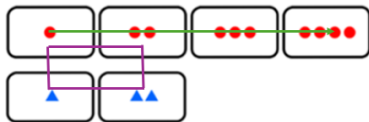


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Maximum Number of Quads, $M(\ell)$, in ℓ Cards

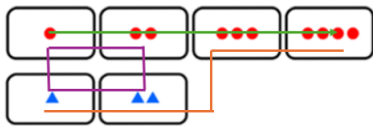
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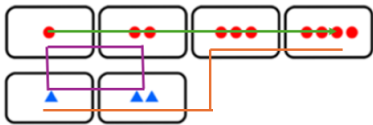
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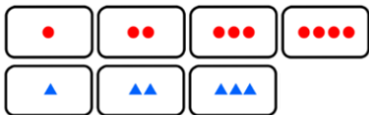
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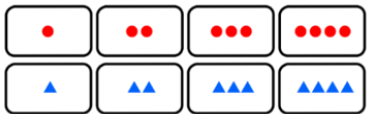
$$M(6) = 3$$

• $\ell = 7$:



$$M(7) = 7$$

• $\ell = 8$:



$$M(8) = 14 = \frac{\binom{8}{3}}{4}$$

Flat-Packing

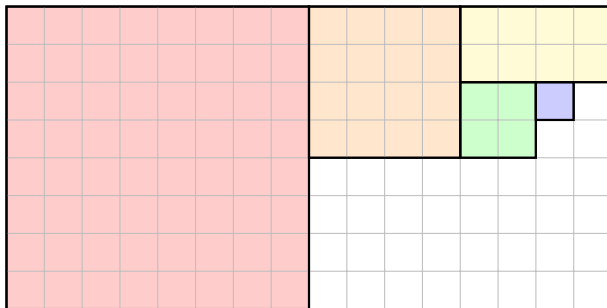
Suppose we want to pack $k = 93$ cards.

$$\begin{aligned} 93 &= 64 + 16 + 8 + 4 + 1 \\ &= 2^6 + 2^4 + 2^3 + 2^2 + 2^0 \end{aligned}$$

Flat-Packing

Suppose we want to pack $k = 93$ cards.

$$\begin{aligned} 93 &= 64 + 16 + 8 + 4 + 1 \\ &= 2^6 + 2^4 + 2^3 + 2^2 + 2^0 \end{aligned}$$



93 cards chosen to form disjoint flats of dimensions 6, 4, 3, 2, 0, all inside a 7-flat. We say that the cards are **flat-packed** with signature (6, 4, 3, 2, 0).

A Formula for Counting Quads

Theorem

Let $\Lambda \subset \mathbb{Z}_2^n$ be flat-packed with signature $\vec{r} = (r_1, r_2, \dots, r_m)$. Then Λ contains precisely $M(\ell)$ quads where,

$$M(\ell) = \sum_{1 \leq i \leq m} \frac{1}{4} \binom{2^{r_i}}{3} + \sum_{1 \leq i < j \leq m} 2^{r_i+r_j-2} (2^{r_i} - 1) + \sum_{1 \leq i < j < k \leq m} 2^{r_i+r_j+r_k-1}$$

ℓ	$M(\ell)$
3	0
4	1
5	1
6	3
7	7
8	14
9	14

ℓ	$M(\ell)$
10	18
11	26
12	39
13	55
14	77
15	105
16	140

ℓ	$M(\ell)$
17	140
18	148
19	164
20	189
21	221
22	263
23	315

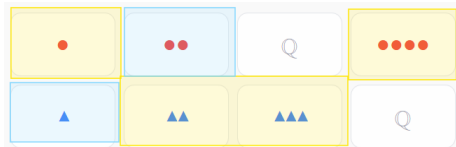
ℓ	$M(\ell)$
24	378
25	442
26	518
27	606
28	707
29	819
30	945

ℓ	$M(\ell)$
31	1085
32	1240
33	1240
34	1256
35	1288
36	1337
37	1401

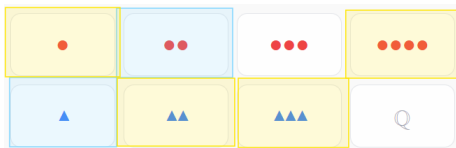
Subsets of a 3-flat with at least 5 cards are flat packed.



- 5 cards, always 1 quad



- 6 cards, always 3 quads



- 7 cards, always 7 quads

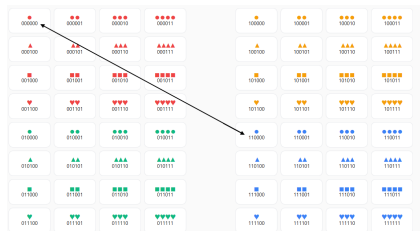
Partitioning and Reflecting

Definition (Reflection Map r)

$\mathbb{Z}_2^n$ is partitioned into two parallel $(n-1)$ -flats, A and B . The function $r_{a_0, b_0}: \mathbb{Z}_2^n \rightarrow \mathbb{Z}_2^n$ defined by $x \mapsto x + a_0 + b_0$ is called a **reflection of $\mathbb{Z}_2^n$ with respect to a_0 and b_0** .

Example

A is L.H.S with $a_0 = 000000$ and B is R.H.S. with $b_0 = 110000$



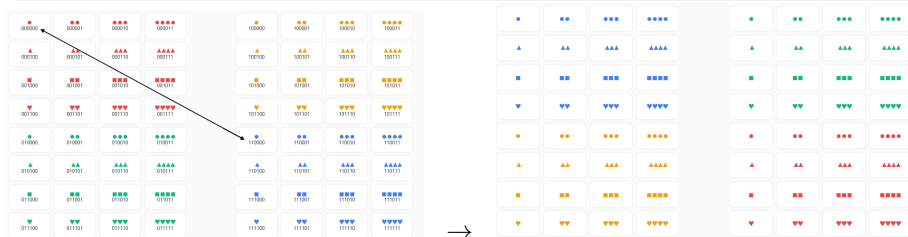
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The φ Map

$\Lambda \subset \mathbb{Z}_2^n$ represents a set of cards. $\Lambda_A = \Lambda \cap A$ and $\Lambda_B = \Lambda \cap B$

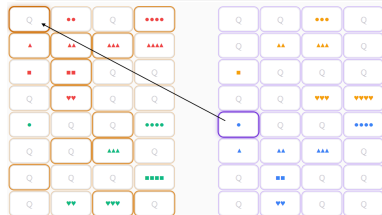
Definition

Define $\varphi: \Lambda \rightarrow \mathbb{Z}_2^n$ by

$$\varphi(x) = \begin{cases} r_{a_0, b_0}(x) & \text{if } x \in \Lambda_B \text{ and } r_{a_0, b_0}(x) \notin \Lambda_A, \\ x & \text{otherwise} \end{cases}$$

Example

A is gold (L.H.S.) with $a_0 = 000000$ and B is purple (R.H.S.) with $b_0 = 110000$



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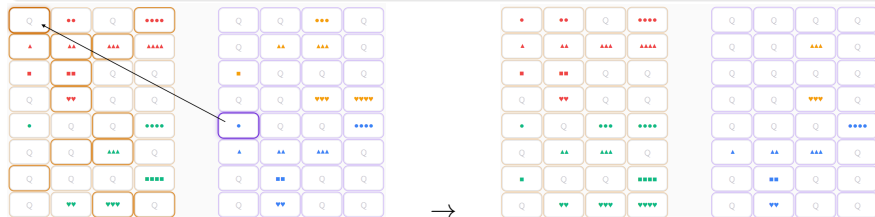
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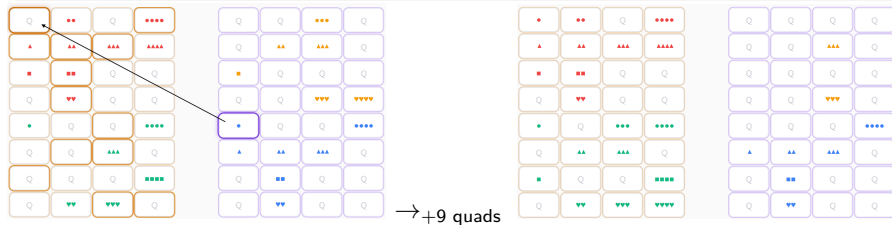
The φ Map Maintains Or Increases the Quad Count

Theorem

$\varphi(\Lambda)$ has at least as many quads as Λ .

Example

A is gold (L.H.S.) with $a_0 = 000000$ and B is purple (R.H.S.) with $b_0 = 110000$
 Λ contains 338 quads while $\varphi(\Lambda)$ contains 347 quads.



Quads that are symmetric under r .

Example

Consider a quad $Q \subset \Lambda$. If $r(Q) = Q$, then Q is fixed under φ .



- Each of these 4 cards is in Λ .
- The φ map used is based on the reflection that pairs blue and red.
- This Λ -quad, Q , reflects onto itself.
- We call Q a symmetric quad

Quads that are asymmetric under r

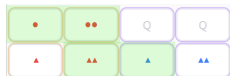
Example

Consider an asymmetric Λ -quad, Q .

- $Q \cup r(Q)$ is a 3-flat
- There are the same number of quads before and after the φ map.
- This example shows 3 quads in the same 3-flat
 - ▶ 1 symmetric on previous slide
 - ▶ 2 asymmetric quads shown here: Q_1 and Q_2

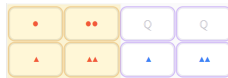


Λ quad Q_1



$\varphi(Q_1)$ not a quad

but



asymmetric Λ' quad.

Quads that are asymmetric under r

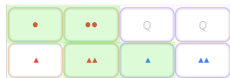
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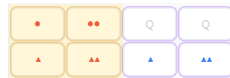


Λ quad Q_1



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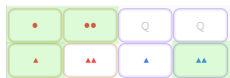
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asymmetric Λ' quad.

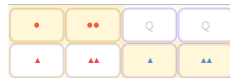


Λ quad Q_2



$\varphi(Q_2)$ not a quad

but



asymmetric Λ' quad.

Outline of proof that φ maintains or increases quads.

Lemma

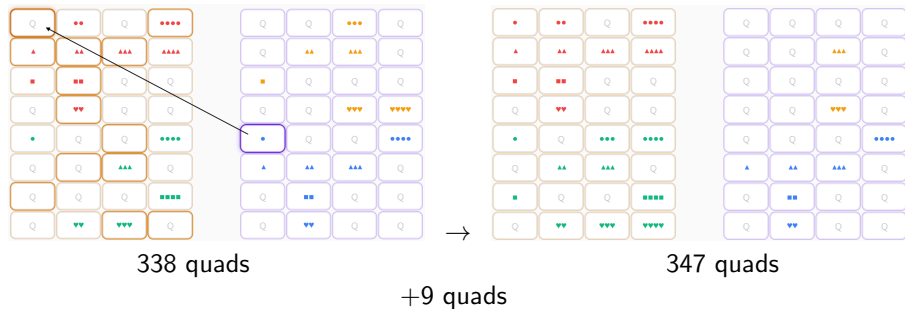
Let $\Lambda' := \varphi(\Lambda)$. There is an injective function $f: \text{quads}(\Lambda) \rightarrow \text{quads}(\Lambda')$.

Proof.

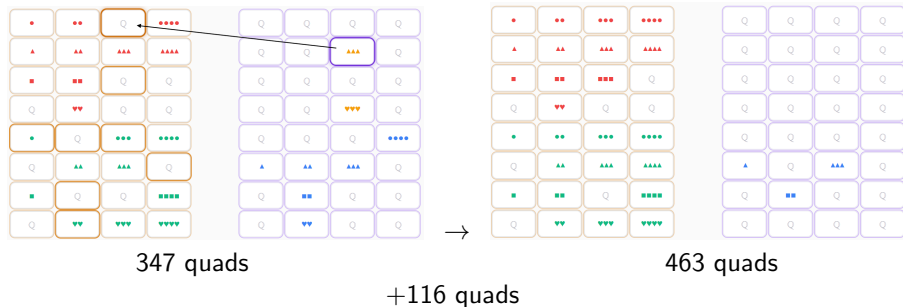
- . Define $f: \text{quads}(\Lambda) \rightarrow \text{quads}(\Lambda')$ piecewise.
 - For $Q \in \Lambda$ s.t. $r(Q) = Q$, then let $f(Q) = Q$.
 - For $Q \in \Lambda$ s.t. $r(Q) \neq Q$, then consider the 3 flat given by $Q3 = Q \cup r(Q)$.
 - ▶ $\varphi(Q3 \cap \Lambda) = Q3 \cap \Lambda'$
 - ▶ $Q3 \cap \Lambda$ and $Q3 \cap \Lambda'$ both contain the same number of cards and at least one quad.
 - ▶ The number of quads is determined by the number of cards. Therefore, $Q3 \cap \Lambda$ and $Q3 \cap \Lambda'$ contain the same number of quads.
 - ▶ This allows one to define f for quads in $Q3 \cap \Lambda$



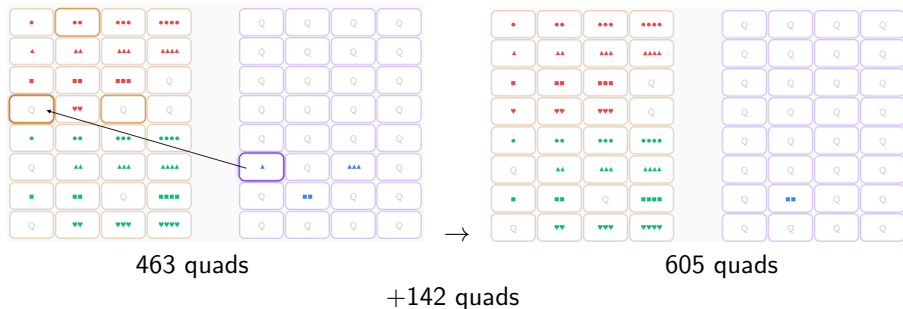
Iterate φ map until a set of cards is flat-packed.



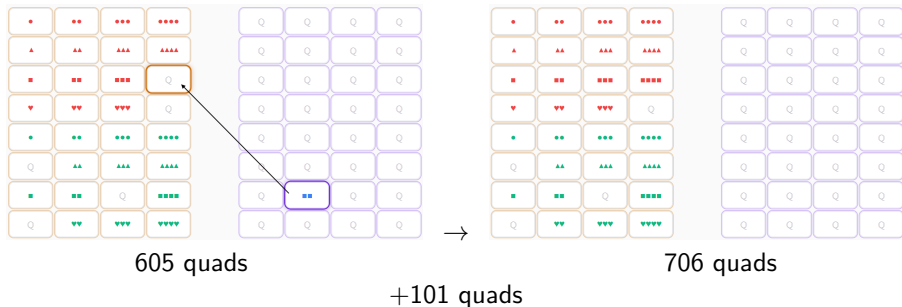
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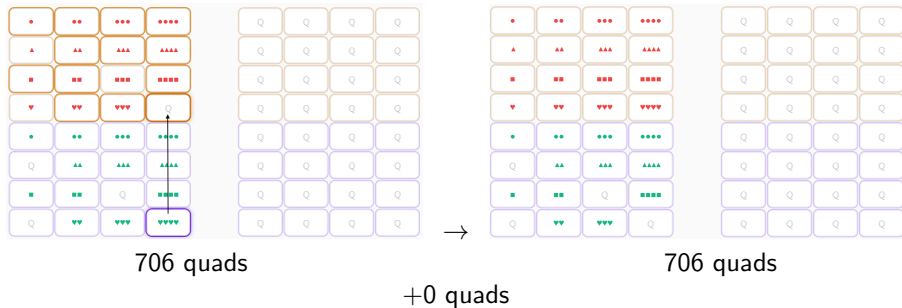
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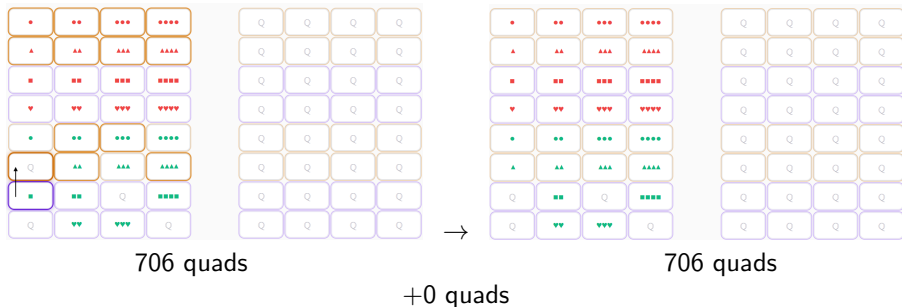
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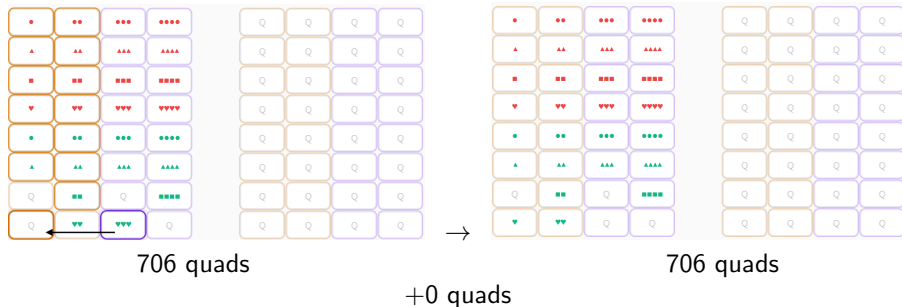
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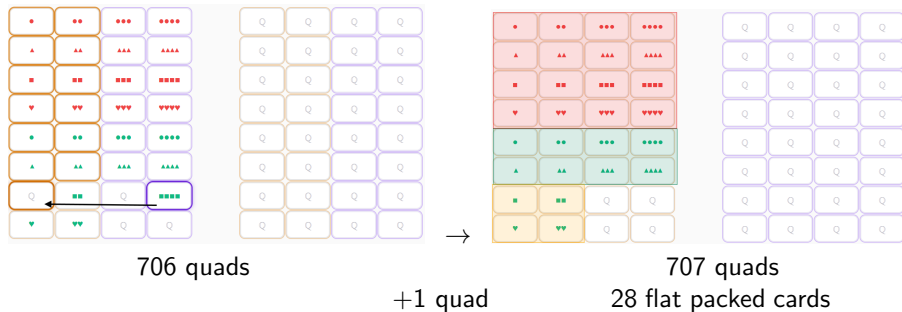
Iterate φ map until a set of cards is flat-packed.



Iterate φ map until a set of cards is flat-packed.

Theorem

A flat-packed set of ℓ cards has the maximum number of quads, $M(\ell)$.



6. Maximal Packing is Flat Packing

- (1) conjectured that if $\dim(\Lambda) > \lceil \log_2(\ell) \rceil$, then the number of quads is at most $M(\ell - 1)$.
- We proved the conjecture using the φ map and combinatorial techniques.

Example

Λ contains 28 cards but Λ spans 6 dimensions. Then Λ contains **at most** $M(27)$ quads.

Results and Conjectures

- **Result 1.** An explicit formula for counting quads in a flat packed set.
- **Result 2.** Flat packing is a max packing.
- **Result 3.** Flat packing is the only max packing.

Quad Packing App

https://math.hope.edu/harrelson/quads/quad_packing.html

- Shows number of quads in a given deck.
- Phi Map: Given cards from opposite flats, performs a φ mapping.
- Find Quads: Interactive activity to find quads in a given deck.

Bibliography

- 1 N. Byrapuram, H. Choi, A. Ge, S. Ge, T. Khovanova, S. Zia Lee, E. Liang, R. Mandal, A. Oki, D. Wu, M. Yang, *Maximum Number of Quads*, Intelligence Planet, 2024.

The End

